A 37-variable quartic equation

Strictly positive integer unknowns; 158 binary-operation abbreviations; exact degree 4.

The 37 unknowns are the following distinct names, each a positive integer:
`argJ`, `argFuel`, `argAlpha`, `argPellF`, `argIndex`, `argTime`, `argX`
`argY`, `argOrdinate`, `argEta`, `argZeta`, `argAuxOrd`, `argGamma`, `argDifferencePlusOne`
`argScale`, `argMainScale`, `argRank`, `argEll`, `argGuard`, `argCube`, `argProduct`
`argFSquare`, `argCSquare`, `argFirstQ`, `argMainQ`, `argVroot`, `argDelta`, `argQSquare`
`argProductX`, `argScaledIndex`, `argQValue`, `argYSum`, `argOffset`, `argGammaProduct`, `argTimeProduct`
`argRankSeed`, `argRootSkew`
Each monospaced G-name below is an abbreviation, not an additional unknown. The fixed integers β, K, μ, ω are defined after the equation.

$\texttt{G001} = \texttt{argJ} \cdot \beta$
$\texttt{G002} = 1 + \texttt{G001}$
$\texttt{G003} = \texttt{G002} \cdot \texttt{G002}$
$\texttt{G004} = \texttt{G002} \cdot \texttt{argQSquare}$
$\texttt{G005} = \texttt{argY} \cdot \texttt{argQSquare}$
$\texttt{G006} = \texttt{G002} - \texttt{argFuel}$
$\texttt{G007} = \texttt{G006} - \texttt{argAlpha}$
$\texttt{G008} = \texttt{argX} \cdot \texttt{argY}$
$\texttt{G009} = \texttt{argEta} + \texttt{argZeta}$
$\texttt{G010} = \texttt{argY} \cdot \texttt{G009}$
$\texttt{G011} = \texttt{argEta} + \texttt{argYSum}$
$\texttt{G012} = \texttt{G011} \cdot \texttt{G011}$
$\texttt{G013} = \texttt{argProductX} + \texttt{argY}$
$\texttt{G014} = 4 \cdot \texttt{G013}$
$\texttt{G015} = 3 + \texttt{G014}$
$\texttt{G016} = \texttt{G013} \cdot \texttt{G013}$
$\texttt{G017} = \texttt{G015} + \texttt{G016}$
$\texttt{G018} = \texttt{G013} \cdot \texttt{G011}$
$\texttt{G019} = \texttt{argX} + \texttt{G018}$
$\texttt{G020} = \texttt{argProductX} \cdot \texttt{argYSum}$
$\texttt{G021} = \texttt{argCube} \cdot \texttt{G001}$
$\texttt{G022} = K \cdot \texttt{G002}$
$\texttt{G023} = \texttt{argX} + \texttt{G022}$
$\texttt{G024} = \texttt{G007} \cdot \texttt{G023}$
$\texttt{G025} = \texttt{argFuel} \cdot \texttt{G002}$
$\texttt{G026} = \texttt{G024} - \texttt{G025}$
$\texttt{G027} = \texttt{argEll} + \texttt{G009}$
$\texttt{G028} = \texttt{G027} + \texttt{argOrdinate}$
$\texttt{G029} = \texttt{G009} \cdot \texttt{argGuard}$
$\texttt{G030} = \texttt{G028} - \texttt{argRootSkew}$
$\texttt{G031} = \texttt{argScale} \cdot \texttt{G030}$
$\texttt{G032} = \texttt{argY} \cdot \texttt{G001}$
$\texttt{G033} = \texttt{argGuard} \cdot \texttt{argCube}$
$\texttt{G034} = \texttt{G031} + \texttt{G032}$
$\texttt{G035} = \texttt{G034} - \texttt{G033}$
$\texttt{G036} = \texttt{argFirstQ} \cdot \texttt{argFirstQ}$
$\texttt{G037} = \texttt{argEll} \cdot \texttt{G027}$
$\texttt{G038} = 1 + \texttt{G037}$
$\texttt{G039} = \texttt{G036} - \texttt{G038}$
$\texttt{G040} = \texttt{G006} \cdot \texttt{G001}$
$\texttt{G041} = \texttt{argAlpha} + \texttt{G040}$
$\texttt{G042} = \texttt{argQSquare} - 1$
$\texttt{G043} = \texttt{argRankSeed} \cdot \texttt{G042}$
$\texttt{G044} = \omega \cdot \texttt{G002}$
$\texttt{G045} = \mu + \texttt{G044}$
$\texttt{G046} = \texttt{argJ} \cdot \texttt{G045}$
$\texttt{G047} = \texttt{G043} + \texttt{G046}$
$\texttt{G048} = \texttt{G009} - \texttt{argRank}$
$\texttt{G049} = \texttt{G048} - 1$
$\texttt{G050} = \texttt{argMainScale} \cdot \texttt{argGamma}$
$\texttt{G051} = \texttt{argProductX} \cdot \texttt{G002}$
$\texttt{G052} = \texttt{argGammaProduct} + \texttt{argProduct}$
$\texttt{G053} = \texttt{G052} - \texttt{G049}$
$\texttt{G054} = \texttt{argMainScale} \cdot \texttt{argOffset}$
$\texttt{G055} = \texttt{G015} \cdot \texttt{G053}$
$\texttt{G056} = \texttt{G054} + \texttt{G055}$
$\texttt{G057} = \texttt{argMainQ} \cdot \texttt{argMainQ}$
$\texttt{G058} = \texttt{argDelta} \cdot \texttt{argCSquare}$
$\texttt{G059} = 1 + \texttt{G058}$
$\texttt{G060} = \texttt{G057} - \texttt{G059}$
$\texttt{G061} = \texttt{argPellF} \cdot \texttt{argPellF}$
$\texttt{G062} = \texttt{argFSquare} - 1$
$\texttt{G063} = \texttt{argDelta} \cdot \texttt{G062}$
$\texttt{G064} = \texttt{argCSquare} \cdot \texttt{argIndex}$
$\texttt{G065} = \texttt{argScaledIndex} \cdot \texttt{argScaledIndex}$
$\texttt{G066} = \texttt{G065} - \texttt{argQValue}$
$\texttt{G067} = \texttt{argTime} \cdot \texttt{G011}$
$\texttt{G068} = \texttt{argTimeProduct} - \texttt{argRank}$
$\texttt{G069} = \texttt{argFSquare} \cdot \texttt{G068}$
$\texttt{G070} = \texttt{G069} - \texttt{G011}$
$\texttt{G071} = \texttt{argAuxOrd} \cdot \texttt{argAuxOrd}$
$\texttt{G072} = \texttt{argVroot} \cdot \texttt{argVroot}$
$\texttt{G073} = \texttt{G071} - \texttt{G072}$
$\texttt{G074} = \texttt{argDifferencePlusOne} - 1$
$\texttt{G075} = \texttt{argQValue} \cdot \texttt{G074}$
$\texttt{G076} = \texttt{G071} - \texttt{G075}$
$\texttt{G077} = \texttt{G076} - 1$
$\texttt{G078} = \texttt{G039} \cdot \texttt{G039}$
$\texttt{G079} = \texttt{G060} \cdot \texttt{G060}$
$\texttt{G080} = \texttt{G077} \cdot \texttt{G077}$
$\texttt{G081} = \texttt{G066} \cdot \texttt{G066}$
$\texttt{G082} = \texttt{G078} + \texttt{G079}$
$\texttt{G083} = \texttt{G082} + \texttt{G080}$
$\texttt{G084} = \texttt{G083} + \texttt{G081}$
$\texttt{G085} = \texttt{G074} - \texttt{G073}$
$\texttt{G086} = \texttt{G085} \cdot \texttt{G085}$
$\texttt{G087} = \texttt{G084} + \texttt{G086}$
$\texttt{G088} = \texttt{argScale} - \texttt{G021}$
$\texttt{G089} = \texttt{G088} \cdot \texttt{G088}$
$\texttt{G090} = \texttt{G087} + \texttt{G089}$
$\texttt{G091} = \texttt{argMainScale} - \texttt{G005}$
$\texttt{G092} = \texttt{G091} \cdot \texttt{G091}$
$\texttt{G093} = \texttt{G090} + \texttt{G092}$
$\texttt{G094} = \texttt{argRank} - \texttt{G047}$
$\texttt{G095} = \texttt{G094} \cdot \texttt{G094}$
$\texttt{G096} = \texttt{G093} + \texttt{G095}$
$\texttt{G097} = \texttt{argEll} - \texttt{G020}$
$\texttt{G098} = \texttt{G097} \cdot \texttt{G097}$
$\texttt{G099} = \texttt{argGuard} - \texttt{G026}$
$\texttt{G100} = \texttt{G099} \cdot \texttt{G099}$
$\texttt{G101} = \texttt{G096} + \texttt{G098}$
$\texttt{G102} = \texttt{G101} + \texttt{G100}$
$\texttt{G103} = \texttt{argCube} - \texttt{G004}$
$\texttt{G104} = \texttt{G103} \cdot \texttt{G103}$
$\texttt{G105} = \texttt{argProduct} - \texttt{G051}$
$\texttt{G106} = \texttt{G105} \cdot \texttt{G105}$
$\texttt{G107} = \texttt{argFSquare} - \texttt{G061}$
$\texttt{G108} = \texttt{G107} \cdot \texttt{G107}$
$\texttt{G109} = \texttt{argCSquare} - \texttt{G012}$
$\texttt{G110} = \texttt{G109} \cdot \texttt{G109}$
$\texttt{G111} = \texttt{G102} + \texttt{G104}$
$\texttt{G112} = \texttt{G111} + \texttt{G106}$

$\texttt{G113} = \texttt{G112} + \texttt{G108}$
$\texttt{G114} = \texttt{G113} + \texttt{G110}$
$\texttt{G115} = \texttt{argScale} \cdot \texttt{argFirstQ}$
$\texttt{G116} = \texttt{G115} - \texttt{G035}$
$\texttt{G117} = \texttt{G116} \cdot \texttt{G116}$
$\texttt{G118} = \texttt{argMainScale} \cdot \texttt{argMainQ}$
$\texttt{G119} = \texttt{G118} - \texttt{G056}$
$\texttt{G120} = \texttt{G119} \cdot \texttt{G119}$
$\texttt{G121} = \texttt{argVroot} - \texttt{G070}$
$\texttt{G122} = \texttt{G121} \cdot \texttt{G121}$
$\texttt{G123} = \texttt{argDelta} - \texttt{G017}$
$\texttt{G124} = \texttt{G123} \cdot \texttt{G123}$
$\texttt{G125} = \texttt{G114} + \texttt{G117}$
$\texttt{G126} = \texttt{G125} + \texttt{G120}$
$\texttt{G127} = \texttt{G126} + \texttt{G122}$
$\texttt{G128} = \texttt{G127} + \texttt{G124}$
$\texttt{G129} = \texttt{argQSquare} - \texttt{G003}$
$\texttt{G130} = \texttt{G129} \cdot \texttt{G129}$
$\texttt{G131} = \texttt{argProductX} - \texttt{G008}$
$\texttt{G132} = \texttt{G131} \cdot \texttt{G131}$
$\texttt{G133} = \texttt{argScaledIndex} - \texttt{G064}$
$\texttt{G134} = \texttt{G133} \cdot \texttt{G133}$
$\texttt{G135} = \texttt{argQValue} - \texttt{G063}$
$\texttt{G136} = \texttt{G135} \cdot \texttt{G135}$
$\texttt{G137} = \texttt{G128} + \texttt{G130}$
$\texttt{G138} = \texttt{G137} + \texttt{G132}$
$\texttt{G139} = \texttt{G138} + \texttt{G134}$
$\texttt{G140} = \texttt{G139} + \texttt{G136}$
$\texttt{G141} = \texttt{argRankSeed} - \texttt{G041}$
$\texttt{G142} = \texttt{G141} \cdot \texttt{G141}$
$\texttt{G143} = \texttt{argYSum} - \texttt{G010}$
$\texttt{G144} = \texttt{G143} \cdot \texttt{G143}$
$\texttt{G145} = \texttt{argRootSkew} - \texttt{G029}$
$\texttt{G146} = \texttt{G145} \cdot \texttt{G145}$
$\texttt{G147} = \texttt{argOffset} - \texttt{G019}$
$\texttt{G148} = \texttt{G147} \cdot \texttt{G147}$
$\texttt{G149} = \texttt{argGammaProduct} - \texttt{G050}$
$\texttt{G150} = \texttt{G149} \cdot \texttt{G149}$
$\texttt{G151} = \texttt{argTimeProduct} - \texttt{G067}$
$\texttt{G152} = \texttt{G151} \cdot \texttt{G151}$
$\texttt{G153} = \texttt{G140} + \texttt{G142}$
$\texttt{G154} = \texttt{G153} + \texttt{G144}$
$\texttt{G155} = \texttt{G154} + \texttt{G146}$
$\texttt{G156} = \texttt{G155} + \texttt{G148}$
$\texttt{G157} = \texttt{G156} + \texttt{G150}$
$\texttt{G158} = \texttt{G157} + \texttt{G152}$

The single equation is $\boxed{\texttt{G158} = 0}$.

This is an external source-mapping candidate. Its zero set has been related to an earlier 40-variable quartic; a Lean proof of the new 37-variable endpoint has not been completed.

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{47}, \quad B = \rho^{2011}, \quad \beta = B - 1,$$
$$G_m(x) = \textstyle\sum_{j=0}^{m-1} x^j, \quad \Lambda = \rho^{1005}.$$

$$\begin{aligned}\mu &= \beta - G_{951}(\rho) \\ &\quad - \rho^{960} G_{26}(\rho) - \rho^{1004}, \\ \omega &= \beta + 8G_{917}(\rho) - 4 + (2^{45} - 4)\Lambda \\ &\quad + \rho^{960} G_9(\rho) + \rho^{987} G_8(\rho) + \rho^{1004}, \\ K &= \rho^{18} + \rho^{1004} + \rho^{2020} \\ &\quad + 2\Lambda\Phi(1/\rho),\end{aligned}$$

where

$$\begin{aligned}\chi_i &= 1 + 2^{10}(1 + w_i + 2^8 n_i \\ &\quad + 2^{17} p_i + 2^{25} s_i) \quad (0 \le i \le 941),\end{aligned}$$
$$\pi(0) = 0, \quad \pi(i) = 1 + \sum_{j=1}^{941} \left\lfloor \frac{2^{44} - 1 + \chi_j - \chi_i}{2^{44}} \right\rfloor \quad (1 \le i \le 941).$$

$$\begin{aligned}\Phi(x) = &\sum_{i=0}^{941} \chi_i x^{\pi(i)} \\ &+ 2^{10} x^{969}(x^9 + 2^{17}) G_8(2x) \\ &+ 2^{18} x^{942}(x^{18} + 2^{17}) G_9(2x).\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 942 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. For $d = 0, 1$, let

$$h(q,b) = 2q + b + 2, \quad \mathcal{S}_d = \{u_j : d_j = d,\ 0 \le j < 430\},$$
$$r_{d,q} = 205 - 199d + |\mathcal{S}_d \cap \{0, \dots, q-1\}|.$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 941$; all index ranges are fixed:

index i	tile $t_i = (w_i, n_i, p_i, s_i)$
0	$(2, 435, 3, 1)$
$j+1$ $(j = 2q + b < 430)$	$((1 - d_j) r_{0,u_j}, h(q,b), d_j r_{1,u_j}, v_j)$
$433 - 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 432a + b)$
$\substack{423+2r_{d,q}+b \\ q \in \mathcal{S}_d,\ d,b=0,1}$	$(dr_{d,q}, b, (1-d) r_{d,q}, h(q,b))$
$923 + 2a + b$ $(a, b = 0, 1)$	$(0, 432a + b, 0, 438)$
927	$(2, 434, 2, 0)$
928	$(3, 436, 4, 4)$
929	$(4, 436, 4, 0)$
$496 + t$ $(t = 434, 435, 436)$	$(1, 438, 1, t)$
$933 + 3a + b$ $(a, b = 0, 1, 2)$	$(a^2, 437, b, 437)$

A closed decoder for the 430 actions. Put $D_b(A,k) = \lfloor A/b^k \rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$\mathcal{N} = \mathcal{A} \bmod 2^{636}, \quad \mathcal{T} = \lfloor \mathcal{A}/2^{636} \rfloor,$$
$$\nu_j = D_2(\mathcal{N}, j), \quad \kappa_j = (\sigma_j + 2)^{1-\nu_j} \quad (0 \le j < 430),$$
$$\sigma_j = \textstyle\sum_{k=0}^{j-1} \nu_k, \quad \Pi_j = \prod_{k=0}^{j-1} \kappa_k \quad (0 \le j \le 430),$$
$$m_q = D_2(\mathcal{N}, 430 + q), \quad \tau_q = \textstyle\sum_{k=0}^{q-1} m_k,$$
$$F_q = 14 + m_q(D_{16}(\lfloor \mathcal{T}/\Pi_{430} \rfloor, \tau_q) - 14),$$
$$u_j = (\sigma_{j+1} - (\lfloor \mathcal{T}/\Pi_j \rfloor \bmod \kappa_j)) \bmod 216,$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums/products are 0/1. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$ in base 10. Its complete decimal expansion is the concatenation of these lines:

```
12889260410359561746736540743515845134244260640012702033212621
95398074180784058571518488282119861023093355389406939414042892
97455773992193784795705001618777025966369137954859764532832722
75411843410809038713502633204617333246733616940964578272360901
56186216508504953649192752154519268024509952082950238865468307
22999632482769253980274091217460707829023970817244790340704449
09085581271192071656170732004937337082171141082486384269072884
72778087419663417942892520175630649792329265834490253035577533
96449689808375788457488971017043689875433162972390177936123032
89025657757489526342497438772045714099661532481123050235898973
27443609903800541199351405985754278307227777584454508422717998
33411356440933534607489268032843 7
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